In [1]:
```
%load_ext autoreload
%autoreload 2

import sys
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import warnings
import joblib
warnings.filterwarnings('ignore')

sys.path.append('../scripts')
from data_pipeline import load_and_prepare_data, load_single_channel, test_t
from feature_engineering import build_state_vector #, normalization (comment
import plots, xgb_scripts, evaluate
```

# Single Cell Model

## State vector

In [2]:
```
df = load_single_channel(
    data_folder="../data/04-03-24",
    channel="A1",
)
df.head()
```

Out[2]:

| | mode | ox/red | error | control changes | Ns changes | counter inc. | Ns | I Range | time/s | control/V |
|---|---|---|---|---|---|---|---|---|---|---|
| **340** | 1.0 | 0.0 | 0.0 | 0.0 | 0.0 | 1.0 | 3.0 | 112.0 | 11100.0 | 80 |
| **341** | 1.0 | 1.0 | 0.0 | 0.0 | 1.0 | 1.0 | 3.0 | 112.0 | 11100.0 | 80 |
| **342** | 1.0 | 1.0 | 0.0 | 0.0 | 0.0 | 1.0 | 3.0 | 112.0 | 11200.0 | 80 |
| **343** | 1.0 | 1.0 | 0.0 | 0.0 | 0.0 | 1.0 | 3.0 | 112.0 | 11200.0 | 80 |
| **344** | 1.0 | 1.0 | 0.0 | 0.0 | 0.0 | 1.0 | 3.0 | 112.0 | 11300.0 | 80 |

5 rows × 43 columns

In [3]:
```
def temporal_split_df(df):
    all_cycles = sorted(df['cycle number'].unique())

    split_index = int(len(all_cycles) * 0.8)
    train_cycles = all_cycles[:split_index]
    test_cycles = all_cycles[split_index:]

    train_data = df[df['cycle number'].isin(train_cycles)]
    test_data = df[df['cycle number'].isin(test_cycles)]

    return train_data, test_data
```

```
df_train, df_test = temporal_split_df(df)

len(df_train), len(df_test)
```

Out[3]: (63442, 16134)

In [4]:
```
X_train, _ = build_state_vector(df_train)
X_test, _ = build_state_vector(df_test)

X_train, X_test
```

```
Out[4]: ([array([ 1.79e-02,  1.78e-02,  1.77e-02,  1.77e-02,  1.76e-02,  1.75e-02,
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          0.00063 ,  0.000408,  0.000115, -0.000274, -0.000796, -0.00147 ,
         -0.00235 , -0.00349 , -0.00495 , -0.00682 , -0.00924 , -0.0122
 ]),
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          0.015   ,  0.0148  ,  0.0146  ,  0.0144  ,  0.0142  ,  0.0139  ,
          0.0137  ,  0.0135  ,  0.0133  ,  0.013   ,  0.0128  ,  0.0125  ,
          0.0123  ,  0.0121  ,  0.012   ,  0.0119  ,  0.0118  ,  0.0118  ,
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         -0.00235 , -0.00349 , -0.00495 , -0.00683 , -0.00924 , -0.0123
 ]),
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          0.015   ,  0.0148  ,  0.0146  ,  0.0144  ,  0.0142  ,  0.014   ,
          0.0138  ,  0.0135  ,  0.0133  ,  0.013   ,  0.0128  ,  0.0126  ,
          0.0124  ,  0.0122  ,  0.012   ,  0.0119  ,  0.0118  ,  0.0118  ,
          0.0119  ,  0.0122  ,  0.0127  ,  0.00101 ,  0.000961,  0.000946,
          0.000961,  0.001   ,  0.00106 ,  0.00112 ,  0.00117 ,  0.00122 ,
          0.00124 ,  0.00123 ,  0.00121 ,  0.00117 ,  0.00112 ,  0.00109 ,
          0.00107 ,  0.00105 ,  0.00102 ,  0.000985,  0.000913,  0.000806,
          0.000654,  0.000431,  0.000141, -0.00025 , -0.000774, -0.00146 ,
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 ]),
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          0.0151  ,  0.0149  ,  0.0147  ,  0.0145  ,  0.0142  ,  0.014   ,
          0.0138  ,  0.0136  ,  0.0133  ,  0.0131  ,  0.0128  ,  0.0126  ,
          0.0124  ,  0.0122  ,  0.012   ,  0.0119  ,  0.0118  ,  0.0119  ,
          0.012   ,  0.0122  ,  0.0125  ,  0.00102 ,  0.000987,  0.000968,
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 ]),
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          0.015   ,  0.0148  ,  0.0146  ,  0.0144  ,  0.0142  ,  0.014   ,
          0.0138  ,  0.0135  ,  0.0133  ,  0.0131  ,  0.0128  ,  0.0126  ,
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            0.00127 ,  0.00126 ,  0.00123 ,  0.00118 ,  0.00113 ,  0.0011  ,
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   ]),
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            0.015   ,  0.0148  ,  0.0146  ,  0.0144  ,  0.0142  ,  0.014   ,
            0.0138  ,  0.0135  ,  0.0133  ,  0.0131  ,  0.0128  ,  0.0126  ,
            0.0124  ,  0.0122  ,  0.012   ,  0.0119  ,  0.0118  ,  0.0119  ,
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            0.0151  ,  0.0149  ,  0.0147  ,  0.0145  ,  0.0143  ,  0.014   ,
            0.0138  ,  0.0136  ,  0.0133  ,  0.0131  ,  0.0128  ,  0.0126  ,
            0.0124  ,  0.0122  ,  0.012   ,  0.0119  ,  0.0119  ,  0.0119  ,
            0.012   ,  0.0122  ,  0.0126  ,  0.00102 ,  0.000993,  0.000991,
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   ]),
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            0.015   ,  0.0148  ,  0.0146  ,  0.0144  ,  0.0142  ,  0.014   ,
            0.0138  ,  0.0136  ,  0.0133  ,  0.0131  ,  0.0128  ,  0.0126  ,
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            0.012   ,  0.0122  ,  0.0127  ,  0.00101 ,  0.000976,  0.000962,
            0.000991,  0.00103 ,  0.00108 ,  0.00115 ,  0.00121 ,  0.00126 ,
            0.00126 ,  0.00125 ,  0.00122 ,  0.00117 ,  0.00113 ,  0.00109 ,
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   ]),
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            0.0153  ,  0.015   ,  0.0148  ,  0.0146  ,  0.0144  ,  0.0142  ,
            0.0139  ,  0.0137  ,  0.0134  ,  0.0132  ,  0.0129  ,  0.0127  ,
            0.0124  ,  0.0122  ,  0.0121  ,  0.0119  ,  0.0119  ,  0.0119  ,
            0.012   ,  0.0122  ,  0.0127  ,  0.00106 ,  0.00104 ,  0.00104 ,
            0.00109 ,  0.00113 ,  0.00119 ,  0.00126 ,  0.00131 ,  0.00136 ,
            0.00136 ,  0.00134 ,  0.0013  ,  0.00125 ,  0.0012  ,  0.00117 ,
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   ]),
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          0.0153  ,  0.0151  ,  0.0149  ,  0.0147  ,  0.0144  ,  0.0142  ,
          0.014   ,  0.0137  ,  0.0135  ,  0.0132  ,  0.0129  ,  0.0127  ,
          0.0125  ,  0.0123  ,  0.0121  ,  0.012   ,  0.0119  ,  0.0119  ,
          0.012   ,  0.0122  ,  0.0125  ,  0.0011  ,  0.00105 ,  0.00107 ,
          0.0011  ,  0.00116 ,  0.00122 ,  0.00131 ,  0.00137 ,  0.00139 ,
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          0.0152  ,  0.0149  ,  0.0147  ,  0.0145  ,  0.0143  ,  0.0141  ,
          0.0139  ,  0.0136  ,  0.0134  ,  0.0131  ,  0.0129  ,  0.0127  ,
          0.0124  ,  0.0122  ,  0.0121  ,  0.0119  ,  0.0119  ,  0.0119  ,
          0.012   ,  0.0122  ,  0.0127  ,  0.00106 ,  0.00103 ,  0.00103 ,
          0.00107 ,  0.00111 ,  0.00117 ,  0.00125 ,  0.0013  ,  0.00134 ,
          0.00135 ,  0.00133 ,  0.00128 ,  0.00122 ,  0.00117 ,  0.00113 ,
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]),
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          0.0139  ,  0.0137  ,  0.0134  ,  0.0132  ,  0.0129  ,  0.0127  ,
          0.0124  ,  0.0123  ,  0.0121  ,  0.012   ,  0.0119  ,  0.0119  ,
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]),
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          0.00139 ,  0.00136 ,  0.0013  ,  0.00123 ,  0.00118 ,  0.00113 ,
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]),
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          0.0138  ,  0.0136  ,  0.0134  ,  0.0131  ,  0.0129  ,  0.0127  ,
          0.0124  ,  0.0123  ,  0.0121  ,  0.012   ,  0.0119  ,  0.0119  ,
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]),
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]),
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          0.0125  ,  0.0123  ,  0.0121  ,  0.012   ,  0.0119  ,  0.0119  ,
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]),
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```

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        0.0127  ,  0.0125  ,  0.0123  ,  0.0122  ,  0.0121  ,  0.0121  ,
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 ]),
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          0.0191  ,  0.0188  ,  0.0185  ,  0.0182  ,  0.0179  ,  0.0176  ,
          0.0174  ,  0.0171  ,  0.0168  ,  0.0165  ,  0.0163  ,  0.016   ,
          0.0158  ,  0.0156  ,  0.0154  ,  0.0153  ,  0.0152  ,  0.0152  ,
          0.0153  ,  0.0156  ,  0.0159  ,  0.00469 ,  0.00514 ,  0.00554 ,
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          0.0017  ,  0.00159 ,  0.00149 ,  0.00137 ,  0.00124 ,  0.00107 ,
          0.000864,  0.000615,  0.000296, -0.000115, -0.000641, -0.00132 ,
         -0.00221 , -0.00334 , -0.00478 , -0.00663 , -0.00899 , -0.012
]),
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          1.94e-02,  1.91e-02,  1.88e-02,  1.85e-02,  1.82e-02,  1.78e-02,
          1.75e-02,  1.73e-02,  1.70e-02,  1.67e-02,  1.64e-02,  1.61e-02,
          1.59e-02,  1.57e-02,  1.55e-02,  1.54e-02,  1.53e-02,  1.53e-02,
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          9.44e-04,  6.82e-04,  3.61e-04, -5.06e-05, -5.91e-04, -1.28e-03,
         -2.17e-03, -3.31e-03, -4.75e-03, -6.61e-03, -8.98e-03, -1.19e-0
2]),
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          0.0222  ,  0.0213  ,  0.0205  ,  0.02    ,  0.0195  ,  0.0192  ,
          0.0189  ,  0.0187  ,  0.0184  ,  0.0181  ,  0.0178  ,  0.0175  ,
          0.0173  ,  0.017   ,  0.0167  ,  0.0165  ,  0.0162  ,  0.016   ,
          0.0158  ,  0.0156  ,  0.0154  ,  0.0153  ,  0.0152  ,  0.0152  ,
          0.0153  ,  0.0156  ,  0.016   ,  0.00426 ,  0.00468 ,  0.00508 ,
          0.00533 ,  0.00538 ,  0.00515 ,  0.00473 ,  0.00419 ,  0.00362 ,
          0.00312 ,  0.00267 ,  0.00234 ,  0.00208 ,  0.00189 ,  0.00175 ,
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]),
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          0.0191  ,  0.0189  ,  0.0186  ,  0.0183  ,  0.018   ,  0.0177  ,
          0.0174  ,  0.0172  ,  0.0169  ,  0.0166  ,  0.0164  ,  0.0161  ,
          0.0159  ,  0.0157  ,  0.0155  ,  0.0154  ,  0.0153  ,  0.0153  ,
          0.0155  ,  0.0157  ,  0.016   ,  0.00446 ,  0.00493 ,  0.00532 ,
          0.00556 ,  0.00557 ,  0.00531 ,  0.00485 ,  0.00428 ,  0.0037  ,
          0.00317 ,  0.00273 ,  0.00238 ,  0.00212 ,  0.00194 ,  0.0018  ,
```

```
         0.00168 ,  0.00157 ,  0.00147 ,  0.00136 ,  0.00122 ,  0.00105 ,
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         0.00578 ,  0.00576 ,  0.00547 ,  0.00497 ,  0.00436 ,  0.00377 ,
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         0.00173 ,  0.00163 ,  0.00151 ,  0.00141 ,  0.00126 ,  0.00108 ,
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         0.0178  ,  0.0175  ,  0.0172  ,  0.0169  ,  0.0167  ,  0.0164  ,
```

```
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         0.0179  ,  0.0176  ,  0.0173  ,  0.017   ,  0.0168  ,  0.0165  ,
         0.0163  ,  0.0161  ,  0.0159  ,  0.0158  ,  0.0157  ,  0.0157  ,
         0.0158  ,  0.016   ,  0.0164  ,  0.00481 ,  0.00531 ,  0.0057  ,
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]),
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         0.0162  ,  0.016   ,  0.0159  ,  0.0157  ,  0.0157  ,  0.0157  ,
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         0.00322 ,  0.00278 ,  0.00243 ,  0.00216 ,  0.00196 ,  0.00181 ,
         0.0017  ,  0.00158 ,  0.00146 ,  0.00134 ,  0.0012  ,  0.00103 ,
         0.000822,  0.000562,  0.000244, -0.000169, -0.000693, -0.00138 ,
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]),
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         1.82e-02,  1.79e-02,  1.76e-02,  1.73e-02,  1.70e-02,  1.67e-02,
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2]),
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2]),
```

```
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          1.00e-03,  7.42e-04,  4.14e-04, -2.31e-05, -5.59e-04, -1.26e-03,
         -2.15e-03, -3.29e-03, -4.74e-03, -6.59e-03, -8.96e-03, -1.19e-0
2])])
```

In [5]:
```python
X_train = np.array(X_train)
X_test = np.array(X_test)

X_train.shape, X_test.shape
```

Out[5]:
```
((213, 66), (54, 66))
```

## Target Var

In [6]:
```python
print(df_train['cycle number'].unique())
```

```
[  1.   2.   3.   4.   5.   6.   7.   8.   9.  10.  11.  12.  13.  14.
  15.  16.  17.  18.  19.  20.  21.  22.  23.  24.  25.  26.  27.  28.
  29.  30.  31.  32.  33.  34.  35.  36.  37.  38.  39.  40.  41.  42.
  43.  44.  45.  46.  47.  48.  49.  50.  51.  52.  53.  54.  55.  56.
  57.  58.  59.  60.  61.  62.  63.  64.  65.  66.  67.  68.  69.  70.
  71.  72.  73.  74.  75.  76.  77.  78.  79.  80.  81.  82.  83.  84.
  85.  86.  87.  88.  89.  90.  91.  92.  93.  94.  95.  96.  97.  98.
  99. 100. 101. 102. 103. 104. 105. 106. 107. 108. 109. 110. 111. 112.
 113. 114. 115. 116. 117. 118. 119. 120. 121. 122. 123. 124. 125. 126.
 127. 128. 129. 130. 131. 132. 133. 134. 135. 136. 137. 138. 139. 140.
 141. 142. 143. 144. 145. 146. 147. 148. 149. 150. 151. 152. 153. 154.
 155. 156. 157. 158. 159. 160. 161. 162. 163. 164. 165. 166. 167. 168.
 169. 170. 171. 172. 173. 174. 175. 176. 177. 178. 179. 180. 181. 182.
 183. 184. 185. 186. 187. 188. 189. 190. 191. 192. 193. 194. 195. 196.
 197. 198. 199. 200. 201. 202. 203. 204. 205. 206. 207. 208. 209. 210.
 211. 212. 213.]
```

In [7]:
```python
y_train = df_train.groupby('cycle number')['Capacity/mA.h'].last().values
y_test = df_test.groupby('cycle number')['Capacity/mA.h'].last().values

y_train.shape, y_test.shape
```

Out[7]:
```
((213,), (54,))
```

In [8]:
```python
y_train, y_test
```

```
Out[8]: (array([4040., 4020., 4000., 3980., 3960., 3940., 3930., 3910., 3900.,
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                3390., 3380., 3390., 3380., 3390., 3390., 3380., 3380., 3370.,
                3370., 3360., 3350., 3350., 3350., 3350., 3330., 3330., 3340.,
                3320., 3310., 3310., 3300., 3300., 3290., 3280., 3280., 3270.,
                3260., 3260., 3250., 3240., 3240., 3230., 3230., 3220., 3210.,
                3210., 3200., 3190., 3180., 3190., 3180., 3150., 3150., 3150.,
                3130., 3130., 3120., 3120., 3100., 3100., 3090., 3080., 3070.,
                3060., 3050., 3040., 3030., 3030., 3010., 3010., 2980., 2980.,
                2960., 2950., 2970., 2930., 2940., 2910.]),
         array([2910., 2890., 2890., 2870., 2890., 2870., 2860., 2850., 2860.,
                2820., 2830., 2820., 2810., 2800., 2800., 2780., 2770., 2760.,
                2760., 2730., 2730., 2730., 2710., 2710., 2690., 2670., 2670.,
                2680., 2630., 2640., 2620., 2600., 2600., 2580., 2580., 2560.,
                2520., 2500., 2510., 2470., 2490., 2490., 2430., 2410., 2390.,
                2380., 2360., 2370., 2330., 2320., 2310., 2280., 2210., 2240.]))
```

In [9]:

```
def scale_data(X_train, X_test, y_train, y_test):
    freq_mean = X_train.mean(axis=0)
    freq_std  = X_train.std(axis=0)
    X_train = (X_train - freq_mean) / freq_std
    X_test  = (X_test  - freq_mean) / freq_std

    y_train = y_train / y_train[0]
    y_test  = y_test / y_test[0]


    return X_train, X_test, y_train, y_test
```

In [10]:

```
X_train_scaled, X_test_scaled, y_train_scaled, y_test_scaled = scale_data(X_

X_train_scaled, X_test_scaled, y_train_scaled, y_test_scaled
```

```
Out[10]: (array([[-0.84991233, -0.83345353, -0.81244243, ..., -1.4288677 ,
                  -2.434389  , -2.20972745],
                 [-0.93876206, -0.92728313, -0.9140872 , ..., -1.65839169,
                  -2.434389  , -1.17528361],
                 [-1.02761179, -1.02111273, -1.01573198, ..., -1.65839169,
                  -2.434389  , -2.20972745],
                 ...,
                 [ 2.79292662,  2.73207124,  2.69430246, ...,  3.39113623,
                   2.66346774,  1.92804791],
                 [ 3.10390068,  3.06047484,  2.94841441, ...,  3.85018422,
                   3.39173298,  2.96249175],
                 [ 2.61522716,  2.54441204,  2.54183529, ...,  2.93208824,
                   2.84553405,  0.89360407]], shape=(213, 66)),
          array([[2.88177635, 2.82590084, 2.79594724, ..., 3.39113623, 3.02760036,
                  1.92804791],
                 [2.65965202, 2.59132684, 2.59265768, ..., 2.70256424, 2.48140142,
                  2.96249175],
                 [2.79292662, 2.73207124, 2.69430246, ..., 3.16161223, 3.20966667,
                  2.96249175],
                 ...,
                 [5.45841853, 5.31238522, 5.18459955, ..., 4.30923221, 3.93793192,
                  2.96249175],
                 [5.72496772, 5.59387402, 5.4387115 , ..., 4.76828021, 4.30206454,
                  2.96249175],
                 [5.59169313, 5.45312962, 5.33706672, ..., 4.30923221, 4.11999823,
                  2.96249175]], shape=(54, 66)),
          array([1.        , 0.9950495 , 0.99009901, 0.98514851, 0.98019802,
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                 0.96039604, 0.95792079, 0.95544554, 0.9529703 , 0.95049505,
                 0.9480198 , 0.94554455, 0.94306931, 0.94306931, 0.94059406,
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                 0.93069307, 0.92821782, 0.92821782, 0.92821782, 0.92574257,
                 0.92574257, 0.92326733, 0.92326733, 0.92326733, 0.92079208,
                 0.92079208, 0.91831683, 0.91831683, 0.91831683, 0.91584158,
                 0.91584158, 0.91336634, 0.91336634, 0.91336634, 0.91336634,
                 0.91089109, 0.91089109, 0.91089109, 0.91089109, 0.90841584,
                 0.90594059, 0.90346535, 0.90594059, 0.90594059, 0.90346535,
                 0.90346535, 0.90346535, 0.9009901 , 0.9009901 , 0.9009901 ,
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                 0.8960396 , 0.8960396 , 0.89356436, 0.89356436, 0.89356436,
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                 0.84653465, 0.84653465, 0.84653465, 0.84653465, 0.84405941,
                 0.84405941, 0.84405941, 0.84405941, 0.84158416, 0.83910891,
                 0.83663366, 0.83910891, 0.83663366, 0.83910891, 0.83910891,
```

```
        0.83663366, 0.83663366, 0.83415842, 0.83415842, 0.83168317,
        0.82920792, 0.82920792, 0.82920792, 0.82920792, 0.82425743,
        0.82425743, 0.82673267, 0.82178218, 0.81930693, 0.81930693,
        0.81683168, 0.81683168, 0.81435644, 0.81188119, 0.81188119,
        0.80940594, 0.80693069, 0.80693069, 0.80445545, 0.8019802 ,
        0.8019802 , 0.79950495, 0.79950495, 0.7970297 , 0.79455446,
        0.79455446, 0.79207921, 0.78960396, 0.78712871, 0.78960396,
        0.78712871, 0.77970297, 0.77970297, 0.77970297, 0.77475248,
        0.77475248, 0.77227723, 0.77227723, 0.76732673, 0.76732673,
        0.76485149, 0.76237624, 0.75990099, 0.75742574, 0.7549505 ,
        0.75247525, 0.75      , 0.75      , 0.7450495 , 0.7450495 ,
        0.73762376, 0.73762376, 0.73267327, 0.73019802, 0.73514851,
        0.72524752, 0.72772277, 0.72029703]),
 array([1.        , 0.99312715, 0.99312715, 0.9862543 , 0.99312715,
        0.9862543 , 0.98281787, 0.97938144, 0.98281787, 0.96907216,
        0.97250859, 0.96907216, 0.96563574, 0.96219931, 0.96219931,
        0.95532646, 0.95189003, 0.94845361, 0.94845361, 0.93814433,
        0.93814433, 0.93814433, 0.93127148, 0.93127148, 0.92439863,
        0.91752577, 0.91752577, 0.9209622 , 0.90378007, 0.90721649,
        0.90034364, 0.89347079, 0.89347079, 0.88659794, 0.88659794,
        0.87972509, 0.86597938, 0.85910653, 0.86254296, 0.84879725,
        0.8556701 , 0.8556701 , 0.83505155, 0.82817869, 0.82130584,
        0.81786942, 0.81099656, 0.81443299, 0.80068729, 0.79725086,
        0.79381443, 0.78350515, 0.75945017, 0.76975945]))
```

# Model

## Train Model

In [11]:
```python
models = xgb_scripts.train_ensemble_model(X_train, y_train)
```

```
Training an ensemble of 10 models...
```

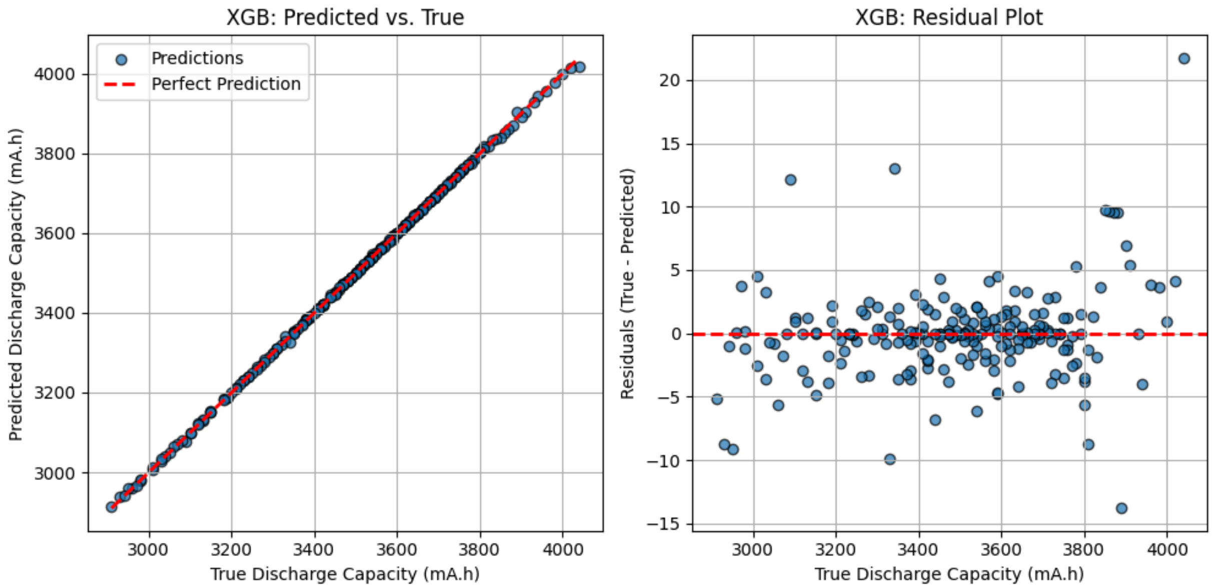
## Confirm Model Predictions

In [12]:
```python
y_pred_mean, y_pred_std, all_predictions = xgb_scripts.predict_ensemble(mode

rmse, r2, mse, mae = evaluate.evaluate_model(y_train, y_pred_mean)
print(f"RMSE: {rmse:.4f}\nR²: {r2:.4f}\nMSE: {mse:.4f}\nMAE: {mae:.4f}")
prediction_fig = plots.model_predictions(y_train.flatten(), y_pred_mean, y_p

plt.show()
```

```
RMSE: 3.5523
R²: 0.9998
MSE: 12.6191
MAE: 2.1264
  Mean Prediction Uncertainty: 4.3895
```



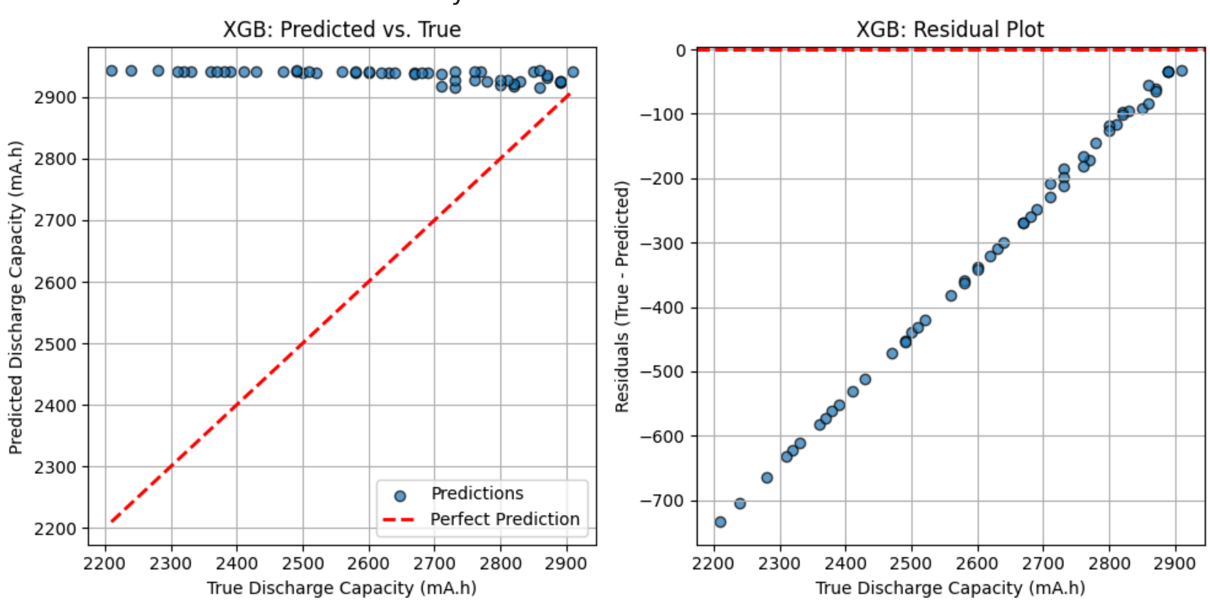

## New Data

In [13]:
```
y_pred_mean, y_pred_std, all_predictions = xgb_scripts.predict_ensemble(mode

rmse, r2, mse, mae = evaluate.evaluate_model(y_test, y_pred_mean)
print(f"RMSE: {rmse:.4f}\nR²: {r2:.4f}\nMSE: {mse:.4f}\nMAE: {mae:.4f}")
prediction_fig = plots.model_predictions(y_test.flatten(), y_pred_mean, y_pr

plt.show()
```

```
RMSE: 367.1798
R²: -2.4841
MSE: 134821.0281
MAE: 306.4781
  Mean Prediction Uncertainty: 8.1942
```




In [14]:
```
print("✅ Notebook ran to completion without errors!")
```

```
✅ Notebook ran to completion without errors!
```